

4(a)	$\rho = m / V$ or $\rho = m / Ah$	B1
	$p = F / A$ or $p = W / A$	B1
	appropriate algebra leading to $p = \rho gh$	B1
	e.g. $p = \rho Ahg / A$ or $\rho Vg / A$ or $\rho Vg / (V/h)$ <u>and</u> (so) $p = \rho gh$	
4(b)	there is atmospheric / air pressure	B1
4(c)	$\Delta p = \rho g \Delta h$ e.g. $(9.66 - 9.60) \times 10^4 / 8.0 \times 10^{-2} = \rho \times 9.81$	C1
	$\rho = 760 - 770 \text{ kg m}^{-3}$	A1
4(d)	$F = \rho g V$ $= 760 \times 9.81 \times 3.7 \times 10^{-4} \times 4.0 \times 10^{-2} (= 0.11 \text{ N})$	C1
	tension = $0.53 - 0.11$ $= 0.42 \text{ N}$	A1
	or	
	$F = (\Delta)p \times A$ $= (9.63 - 9.60) \times 10^4 \times 3.7 \times 10^{-4} (= 0.11 \text{ N})$	(C1)
	tension = $0.53 - 0.11$ $= 0.42 \text{ N}$	(A1)